

How AI Could Benefit Workers, Even If It Displaces Most Jobs

AI is already taking jobs, but that is only one facet of its complex economic effects. Price dynamics and bottlenecks indicate that automation could be good news for workers — but only if it vastly outperforms them.

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Last week, Jack Dorsey, a co-founder of Twitter, announced that his company Block is cutting its head count from 10,000 to fewer than 6,000 because AI tools mean it needs fewer workers. It is not the first company to make such an announcement, and won't be the last. But it raises a question: if AI takes jobs, are workers doomed? To many observers, the answer must be yes. Negative consequences for the labor force seem like an inevitable byproduct of advancing AI. Indeed, machines that automate work seem to promise exactly such outcomes: they start performing tasks that labor once did, which seems to imply that workers will experience worse economic prospects. Faced with the possibility of being displaced, many might hope that AI progress will slow or even stall, allowing humans to remain competitive.

But, in evaluating AI's implications for the workforce, it's misleading to think of AI as simply a replacement for labor. To complete the picture, we need to consider the fuller set of forces that are unleashed when machines automate types of work. In particular, we must engage two other economic features that sharply condition outcomes for labor: first, how machines affect prices, and second, the role of economic

bottlenecks. Analyzing these dimensions leads us to a counterintuitive notion: the labor side of the economy will be far better off economically if AI is much, much better than humans at the jobs it replaces. In other words, rather than rooting for AI to fail, we should hope it exceeds us by a large margin.

Two clarifications before we begin. First, this argument does not assume a near- or mid-term future in which AI replaces all human labor (though we will discuss that possibility at the article's end). It assumes uneven technological progress, in which dramatic productivity gains appear in some domains — and perhaps very many — while others remain more dependent on human labor. Second, when evaluating outcomes for workers, even if average living standards rise substantially, the transition can bring significant disruption, as well as real hardship for some workers. We will discuss this key issue as well.

Before turning to the mechanisms that drive value toward labor, it is useful to ground the discussion in history — specifically, the rise of agricultural machinery and the emergence of computers.

Automating Agriculture and Computing

Automation caused a huge drop in the percentage of workers employed in agriculture. In the United States today, less than 2% of the workforce is engaged in agriculture. But back in the 18th century, agricultural employment occupied 80% to 90% of the US workforce — i.e., almost everybody. This shift from the large majority of workers being farmers to a tiny minority being farmers is typical among leading economies, including France, Germany, Italy, Spain, the UK, and Japan.

So let's imagine you traveled back in time to colonial New England or rural France and told farm laborers: "Look, machines are coming that will replace almost all of your jobs. They'll plant, water, fertilize, harvest, thresh the wheat, shuck the corn, mill the grains..." You might mention a specific machine called a "combine harvester" that in the year 2001 would process 1 million pounds of corn in eight hours. To the extent the farmers believed you, they might naturally have two immediate reactions. First, they would expect farm labor (i.e., pretty much all labor at the time) to become impoverished as the machines took over nearly all of their jobs. Second, and related, they might anticipate that those who owned the farming machines would capture most of the income.

But history unfolded in a strikingly different way, with workers ultimately sharing in unprecedented prosperity. Real income per worker in the US today is about 25 times larger than it was in the late 18th century. Labor continues to capture about two-thirds of total national income. And, while companies like John Deere do very well making and selling combine harvesters and other amazing agricultural machines, their market capitalizations and sales are tiny portions of the overall economy.

Now, AI might not trace the same path as agriculture. But the farming experience should at least open one's mind to the idea that "machines replace labor" is not the

whole economic story, and that it is possible to destroy most of the jobs in an economy and see workforces thrive over the long run.

Computers have become ubiquitous, but they receive only a small fraction of spending. A more current example is computing itself. Over the past 50 years, we have witnessed mind-boggling advances in computer productivity that outpace human capabilities at an ever-expanding array of tasks. Computer use has expanded from niche applications on expensive mainframes, in the 1960s, to ubiquitous devices seen at work, at home, in our pockets, and in machines throughout the economy. Thanks to their ever-expanding capabilities, computers have replaced or substantially infiltrated many types of work — including clerical, secretarial, and manufacturing labor. Reflect for a moment on your “screen time” as a measure of how pervasive computers have become in our lives.

Yet, despite this expansion, expenditures on computing remain modest: business investment in computer equipment and software now peaks at around 4% of GDP. Similarly, households devote tiny shares of their budgets to computers and software — we spend many multiples more on housing, transportation, health care, or even restaurant meals.

How can it be that computers have taken over our lives, and yet we spend such small shares of our income on them? And how did farming machinery replace the vast majority of jobs in the economy, even as labor continued to thrive? To help answer these questions, we need to think about how these machines affect prices and about the knock-on effects for how value is distributed across economic sectors.

Machines and Prices

Computers are so efficient that the price of computation is close to zero. Computers represent a small share of the economy because, although they perform a huge quantity of tasks, they do so at very low cost. The result is that the services they provide become very inexpensive. In fact, many computer services are provided to consumers virtually for free. Housing is not free. Haircuts are not free. But internet search is effectively free. Offering things for free is partly about business models, but it is largely about how efficient the technology is. For example, an average human can do a simple mathematical calculation — say multiplying two four-digit numbers — in about one minute. My smartphone can do about 500 billion such calculations in one minute. Comparing the median wage to the cost of electricity, my smartphone is also about 5,000 times less expensive than a person over that one minute. So computers make multiplication virtually free.

The better a machine is at automating a task, the lower it will drive prices for consumers. This brings us to the first, major edit to the idea that machines replacing labor is necessarily bad for the labor force. Because automation doesn’t just replace work. It also makes output cheaper. In fact, that is the exact reason we deploy automation in the first place. Businesses, in pursuit of profits, generally seek to produce their goods and services at lower cost, so they use whichever approach — with whatever combination of machines or humans — they believe to be cheapest.¹ On

the other side, consumers typically prefer lower prices for a given product or service, driving demand toward the firms that are more cost-efficient. Indeed, widespread discontent over inflation in the US reminds us that people care a whole lot about prices, not just jobs, when assessing their standard of living.

If AI takes some jobs, let's hope it excels at them. Following this logic, we can see that the more productive the machine, the lower prices will fall, and the more consumers will benefit. Thus, if a machine like an AI proves the better choice over human effort for any given task, then we should hope it is way, way better than we are at that task. If so, its output gets really cheap.

Of course, even if goods and services get much cheaper, people will still need a source of income in order to afford them. At this point, we have a tradeoff — automation makes things less expensive, but it also replaces human work. So how can the labor force win, in the balance? This brings us to the second critical feature of modern economies: bottlenecks.

Bottlenecks

Let's return to farming and consider the price of corn. With the help of machines like combine harvesters, the price of corn at the farm gate is now only about 10 cents per pound of kernels. Yet cornflakes, corn syrup, and corn chips cost 20 to 40 times that (e.g., \$4 for a one-pound bag of corn chips). This is not because the farm or some other business is charging giant markups over its corn costs. It is because there are many other steps and associated costs downstream of the farm, including transportation, storage, processing, packaging, and retail services. Each of these currently requires many labor tasks, many of which remain hard and unproductive. There is no “amazing machine” for stocking shelves, building the grocery store, or fixing a flat tire when a truck breaks down.

Tasks that cannot be automated end up receiving a large share of economic value. Here's the critical and perhaps surprising result: the economy is largely a story of what we do badly, not what we do well. This follows the same logic whereby amazing machines make their output cheap. The related implication is that the worse we are at something, the more expensive it will be. The expensive parts of the economy tend to be the things we need to do but haven't figured out how to improve. And these so-called bottlenecks are everywhere. Combine harvesters have made harvesting corn the wide part of the bottle. Getting corn from the farm to a corn chip in your hand? Therein lie many tasks that constitute the narrower part of the bottle. And that's where the payments go.

Due to uneven productivity gains, we spend more on restaurant meals than computers. This phenomenon — that we end up spending most of our time and money on the things we are unable to automate — is known to economists as “Baumol's cost disease.” It helps explain why agriculture and manufacturing have been declining as a share of US GDP, while services — education, health care, government, finance, insurance, transportation — take over larger shares of GDP. For example, in a restaurant, the productivity of chefs, prep cooks, servers, bartenders,

and dishwashers has made at most modest advances over many decades. Today we spend several times more on restaurant meals than on our ubiquitous computers.

Despite AI tools for medical diagnosis, health care delivery remains labor-constrained. AI systems are already showing impressive capabilities in medical-imaging analysis, diagnosis, and administrative documentation. As these tools improve, the cost of clinical decision-making is likely to fall sharply. But cheaper and better medical assessment does not reduce the need for care. And the delivery of care itself remains constrained by human labor. Nurses, physicians, technicians, caregivers, and support staff have many roles to play. Hospitals must still be staffed. Treatments must still be administered; surgeries must be performed.²

According to this logic, successful automation does not eliminate the need for human work. Rather, it shifts the demand for labor, concentrating value and wages in the parts of the economy where human time remains the limiting factor.

Price dynamics and economic bottlenecks give us good reasons to be hopeful for labor. If labor continues to perform various bottleneck tasks, while machines get very good at everything else, we end up in a situation where labor does very well. In fact, let's take it to the limit. Let's say AI becomes so efficient that it can do tasks like writing computer code at close to zero cost compared with a human performing the same assignment. And let's say AI advances similarly at a large share of other tasks. Then, what we pay AI and related machines, as a share of GDP, becomes very small indeed. And the share of total income that goes to workers? It goes up.

Three Caveats

The above arguments can help explain major historical transformations and show how advanced AI could be advantageous for workers. But there are several caveats to acknowledge, where the economic prognosis may become less rosy.

Marginally better AI

AI that only marginally outperforms humans could be a net negative for workers. Today's plentiful food supply and diversified economy lean on machines that are incomparably more efficient than we are at many forms of agricultural work. But imagine if such tools had instead offered only modest productivity gains over human labor. Such "marginally better" machines could have led to lost jobs without meaningfully lowering food prices for consumers. Moreover, because the amount spent on the machines in such an economy would be high, they would take over a larger share of GDP, with labor getting paid a smaller share. This is exactly why we want AI to excel at what it takes over, not just provide a slight improvement.

That said, there are natural reasons to think that AI productivity gains will be more than slight. First, computers tend to be fast, producing a lot of calculations in a short period. Second, computers don't cost very much per unit of time, primarily because electricity usage is less expensive than wages. Where AI is able to perform a task successfully, it seems likely to lower costs considerably.

Rapid job loss and a lack of alternative work

For jobs that are replaced, workers' outcomes hinge on whether they can transition into new roles. A second — and potentially profound — concern is whether displaced workers will be able to find new employment. In flexible labor markets, workers can move toward bottlenecks, performing roles in which they may ultimately be better off. But the new job may be inferior to the old job for a given worker, especially if that person was especially skilled at the prior work. And job transitions themselves — which can involve financial strain, loss of identity, and household stress — are painful.

A major factor affecting whether workers can shift to alternative jobs is the speed of automation. Gradual change allows adjustment; rapid displacement can overwhelm workers and communities.

Abrupt automation can carry heavy social costs. History offers a vivid illustration. England's Captain Swing riots, in 1830, followed the rapid adoption of mechanical threshing machines. These machines eliminated a key source of winter employment (grueling though it was) for farm laborers. As the machines spread, laborers rioted. They destroyed equipment, burned buildings, and threatened violence against parish authorities and landowners. This episode shows both the hardship workers can endure as a byproduct of automation and the broader social conflict that it can spark.

Alternative work options and political conditions are critical. Tellingly, areas of England with greater alternative employment opportunities saw few if any riots. Further, the unrest unfolded in a context of weak political rights, limited social support, and extreme poverty. Over the previous decades, England had been reallocating common land to large local landowners — so-called enclosure — creating a large class of landless farm workers who became dependent on wage work. Further, these workers could not vote. Thus deprived of property rights and political rights, they predictably revolted. The lesson, then, is not that automation inevitably leads to turmoil; it's that having access to alternative opportunities and civic power both matter enormously to the social response. This insight informs how we might confront more extreme technology scenarios, as well as our third caveat: the prospect of full automation.

Full automation

What happens if AI exceeds human capacities at all tasks? Full automation of human work strikes me as still far off, because it requires AI that excels at all cognitive and all physical tasks, and it also requires that humans will want to be served by robots for virtually everything we care about. Count me as skeptical that this happens anytime soon. But let's say that we do reach a future when humans can be fully replaced and no bottleneck tasks remain. What happens to incomes then?

Prices might drop close to zero, but workers might receive only a tiny fraction of value. If machines do everything, then those who own the machines will capture all this value. Products and services would become very cheap, but workers, outcompeted

by machines in all tasks, would end up with a vanishingly small share of the economy's income.³

Political systems would need to decide how to distribute machines' vast output. The full automation scenario is so far beyond human experience that we should be careful in making any claims about it. But depriving the large majority of people in society of a means to support themselves, while a few people become incredibly wealthy, does not seem a stable situation. Returning to England in the 1830s: as the Swing riots progressed, the wealthy realized their own personal risk at the hands of workers with little left to lose. The social unrest intensified pressure for political change, contributing to the Great Reform Act of 1832, which began to extend voting rights and bring formal political power to England's middle class. In short, when technological progress pushed workers into an untenable situation, people rose up, and political institutions shifted, working (if imperfectly) to accommodate these challenges. Similarly, political systems will be the essential arbiters in a world of full automation. While full automation would create vast output gains, it could create dystopia or utopia, economically, depending on how we share its fruits.

Labor will fare best if AI automates with extraordinary competency. These caveats matter. They remind us that technological progress is not automatically benign, that transitions can be painful, and that institutions shape outcomes. But it is also easy to let the most dramatic possibilities dominate our thinking. The future is unlikely to jump directly to dystopia or utopia. More plausibly, AI will advance unevenly — extraordinarily powerful in some domains, limited in others. And, in a world of profound yet uneven transformation, the fate of workers will again hinge on the same two questions that shaped agriculture and computing: how cheap automation makes what it does well, and where the narrow parts of the bottle remain. The promise of AI, then, is not that it preserves existing jobs but that, by collapsing prices in the tasks it masters, it pushes value to the tasks it does not master. So, if AI is going to beat us at a task, let's hope it beats us by a mile. Then, ironically, labor may yet win.

FOOTNOTES

1. A given business may get this wrong in the short run, but the businesses that get this right can produce the same good or service at lower cost, and hence they can charge less than other firms, attracting buyers and winning the market.
2. Are we close to a world where robots are intubating patients, implanting IV lines, establishing sterile fields, managing emergency bleeding, administering CPR, changing bedsheets, moving patients around hospitals, or conducting insurance appeals? To me, this seems like a far-off prospect.
3. Interestingly, even here, standards of living could still (broadly) go up. Humans could still work, but we would now be competing with machines at any task. The market price of output would decline. For example, let's say you produce shoes at a certain rate per hour, while a cheap machine produces many more shoes per hour. You can still make shoes; the problem is that the price you can sell shoes for will drop significantly, so you will earn less as a shoemaker. At the same time, the price of other things in the economy will also be low, because machines are similarly taking over those activities. So, although you make less as a shoemaker, you can still do quite well at turning an hour of your work into the things you buy. This is a version of

the Ricardian gains from trade, where you can gain from trade even if you don't have an absolute advantage at any task. So the economic threat of a full-automation scenario is less that humans' standard of living will go down (in an absolute sense) and more that the world will become extraordinarily unequal, with the gains from all these machine advances accruing to the machine owners.

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